

“Embark the Journey · Embrace the Universe”: An Interactive System of Astronomy Education Based on Spherical TUI in MR

Xiaozhan Liang
School of New Media Art and Design,
Beihang University, Beijing, China;
State Key Laboratory of Virtual
Reality Technology and Systems,
Beihang University, Beijing, China
dearzhan1019@163.com

Yong Hu
School of New Media Art and Design,
Beihang University, Beijing, China;
Yunnan Innovation Institute of
Beihang University, Yunnan, China
huyong@buaa.edu.cn

Xukun Shen
School of New Media Art and Design,
Beihang University, Beijing, China;
State Key Laboratory of Virtual
Reality Technology and Systems,
Beihang University, Beijing, China
xkshen@buaa.edu.cn



Figure 1: (a) Visualization (b) Image overlay (c) Information supplementation (d) Visual cues (e) Model combination

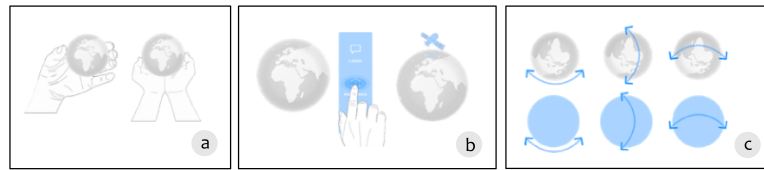


Figure 2: (a) Direct interaction on the sphere (b) Direct interaction around the sphere (c) Mapping interaction

ABSTRACT

The widespread adoption and application of MR technology have opened up new possibilities in the field of education. Integrating physical objects and virtual contents in the physical interactive interface provides a larger practical platform for astronomy science education. However, very few attempts are currently at this form of science education. Therefore, we have summarized the design principles for the spherical TUI regarding both interactive interfaces and behaviors. Based on this, we have developed a spherical TUI-based astronomical knowledge system. It delivers users with a more diverse knowledge presentation by a fusion of virtual and reality and mapping interaction of spherical TUI in MR.

ACM Reference Format:

Xiaozhan Liang, Yong Hu, and Xukun Shen. 2023. “Embark the Journey · Embrace the Universe”: An Interactive System of Astronomy Education Based on Spherical TUI in MR. In *SIGGRAPH Asia 2023 XR (SA XR '23)*, December 12–15, 2023, Sydney, NSW, Australia. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3610549.3614609>

1 INTRODUCTION

Mixed reality devices provide a rich, interactive, and immersive environment for education by leveraging tracking and positioning

as well as input interaction mechanisms. Among these mechanisms, gesture is the most commonly used interaction method in MR. However, the lack of haptic perception in the gesture-only method limits the user’s experience. This creates an opportunity for TUI to be applied.

For interactive objects, TUI requires physical objects that are trackable, and easy to operate. The sphere meets these conditions on the basis that it has interactive continuity and operational universality. Furthermore, there is potential for using spherical TUI as celestial bodies in astronomy education. Our contributions are as follows:

- We have summarized the design of interactive interfaces and behaviors for spherical TUI, which can be applied as guidelines in other designs.
- We have applied the results in astronomy education and developed the system (See Figure 3), enriching education forms.

2 DESIGN OF SPHERICAL TUI IN MR

The spherical TUI has been explored in many applications of human-computer interaction (HCI). For example, [[Yamashita et al., 2007]] [[Satriadi et al., 2022]] [[Englmeier et al., 2020]] have used globes as TUI, which inspired our work.

2.1 Interactive Interface Design

Based on the curved features of the sphere, its interface can be designed by displaying visualization diagrams along the exterior (See Figure 1, a). Another form is that spatial anchors can project digital textures on the sphere’s surface (See Figure 1, b).

Based on the application of spheres in astronomical scenarios, the interface can present images next to the sphere to complement it (See Figure 1, c). In addition, the interface can be designed with virtual labels placed on the surface of the sphere (See Figure 1, d). A further interface can form new content by adding virtual models such as the orbits and associating them to spheres in terms of their location relationships (See Figure 1, e).

2.2 Interactive Behavior Design

In the design of direct interaction, movement, touch, and rotation are natural interactions based on the user's perception of spheres (See Figure 2). In addition, it can include cue-based interactions with the space around the sphere.

In the design of mapping interaction, the user can change the position, pose, and shape with the 6 DOF tracking technology to induce changes in the virtual sphere in a continuous form.

3 DESIGN AND IMPLEMENTATION

"Embark the Journey · Embrace the Universe" is an astronomical knowledge system based on the sphere TUI in MR, fusing virtual and real space. The entity of the system uses 3D printing technology. Meanwhile, an MPU6050 gyroscope sensor is embedded in the sphere to obtain attitude data (See Figure 4). Based on this, Wi-Fi communication between the sensor and HoloLens 2 enables real-time mapping.



Figure 3: Exhibition environment scene



Figure 4: Overall system design



Figure 5: The virtual and reality fusion of the landing



Figure 6: Rotational mapping of information content

3.1 The Fusion of Virtual and Reality

In this part of the virtual-real fusion, users can trigger the interactive interface of "Explore the Moon" and "Explore Mars" respectively by tracking the menu of the hand next to the sphere.

Users can interact with virtual content in each part to learn more about aerospace history. During the "Lunar Journey", users can learn about the achievements of Chang'e Project based on multi-modal interaction, such as eye movement, pinching, and button controlling (See Figure 5). In the "Mars" part, users can understand the key missions of "Tianwen-1".

3.2 Mapping Interaction of Spherical TUI

In the part of mapping interaction corresponding to the "Embrace the Universe", users can manipulate the Earth sphere based on its real-time mapping.

Users can select a planet and rotate the sphere at will, the virtual planet will be synchronized (rotation speed and direction). Furthermore, the system also maps the rotation of its 3 axes to 3 pieces of information: "orbits," "components," and "temperature" (See Figure 6).

4 CONCLUSION

This study proposes design rules for the interaction interface and behavior of the sphere in MR and delivers information about space journeys and planets in the solar system through the design practice. Our design work offered new ideas for developing similar teaching and exhibitions.

ACKNOWLEDGMENTS

This work is supported by BeiHang University Yunnan Innovation of Institute Yunding Technology Plan(2021) of Yunnan Provincial Key R&D Program(202103AN080001-003). The authors sincerely thank ring all reviews of their kind suggestions.

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